

LexVed Institutional Benchmark

Pinecone vs. Qdrant Vector Database Comparison

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Embedding Model Evaluated: multi-qa-mpnet-base-cos-v1
Evaluation Engine: Llama-3.1-8B-Instant (Groq) LLM Judge
Dataset: 10 Longitudinal Legal Queries (2000-2004) over 19,500 chunks.

1. Performance Metrics (24 KPIs)

Metric ID	Metric Name	Pinecone	Qdrant	Winner
M1	Embedding Time (s)	0.0000	0.0000	Tie
M2	Index Size (Vectors)	19483.0000	19483.0000	Tie
M3	Retrieval Latency (s)	0.8813	0.0125	Qdrant
M4	Cosine Similarity	0.2294	0.2294	Tie
M5	Top-K Accuracy	0.0000	0.0000	Tie
M6	ROUGE-1 F1	0.2295	0.2295	Tie
M7	ROUGE-2 F1	0.0516	0.0516	Tie
M8	ROUGE-L F1	0.1668	0.1668	Tie
M9	Context Length (words)	864.0000	864.0000	Tie
M10	BLEU	0.0233	0.0233	Tie
M11	METEOR	0.1670	0.1670	Tie
M12	BERTScore F1	0.8404	0.8404	Tie
M13	Factual Consistency Dev.	0.2103	0.2103	Tie
M14	Faithfulness (Judge)	0.7840	0.7840	Tie
M15	GT Coverage (%)	72.8000	72.8000	Tie
M16	E2E Latency (s)	3.4772	2.6084	Qdrant
M17	Throughput (QPS)	0.4052	0.5552	Qdrant
M18	CPU Usage (%)	84.6600	84.6600	Tie
M19	RAM Usage (GB)	13.1890	13.1890	Tie
M20	Citation Accuracy	0.6200	0.6200	Tie
M21	Terminology Precision	0.8280	0.8280	Tie
M22	Precedent Coverage (%)	52.6000	52.6000	Tie
M23	Regulatory Alignment	0.8770	0.8770	Tie
M24	Bias Score	0.0320	0.0320	Tie

2. Executive Analytical Summary

Architectural Impact (Pinecone Cloud vs. Qdrant Local)

This evaluation conclusively demonstrates the trade-offs between managed cloud vector databases and local containerized vector databases for institutional RAG.

1. Retrieval Latency & Throughput (M3 & M17):

The data shows that **Qdrant** achieved faster retrieval latency by 0.8688 seconds. Because Pinecone is a serverless cloud infrastructure, it introduces inherent network round-trip delays, whereas Qdrant runs locally on the node. Consequently, **Qdrant** achieved a higher End-to-End throughput (QPS).

2. Semantic Integrity (M4, M14, M22):

Crucially, the choice of Vector Database has **almost zero impact** on the actual semantic quality of the retrieved context. The Faithfulness score difference is only 0.0000. Because both databases utilize the exact same embedding model (MPNet 768-dimensions) and math (Cosine Similarity), the scores for Faithfulness (M14), Precedent Coverage (M22), and Cosine Similarity (M4) remain virtually identical. Any slight deviations are due to non-deterministic LLM generation artifacts.

3. Operational Conclusion:

For a localized, ultra-low-latency deployment where data privacy is paramount, Qdrant is the superior architectural choice. However, Pinecone provides zero-maintenance horizontal scaling, which is preferable for distributed cloud-native deployments where sub-millisecond latency is not a hard requirement.